

ECON2050 Tutorial 6

Linear Algebra II

Semester 2 2026

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14 & 17 September

Today's Plan

- ▶ Tutorial questions: Q1–Q4

Tutorial Question 1

Problem 1

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 0 & -1 \\ 1 & 1 & 2 \end{pmatrix}.$$

- (i) Sketch the cone generated by the columns of $\mathbf{A}$.
- (ii) What is the cone generated by just the first and third columns of $\mathbf{A}$?
- (iii) If $\mathbf{b} = (1, 0)$, decide whether $\mathbf{Ax} = \mathbf{b}$ has a solution with $\mathbf{x} \geq \mathbf{0}$.

Recap: Span versus cone

Span

$$\text{span}(\mathbf{A}) = \{\mathbf{y} \in \mathbb{R}^m : \mathbf{y} = \mathbf{A}\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^n\}.$$

The coefficients in a span may have any sign.

Cone

$$\text{cone}(\mathbf{A}) = \{\mathbf{y} \in \mathbb{R}^m : \mathbf{y} = \mathbf{A}\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}_+^n\}.$$

The coefficients in a cone must be non-negative.

$$\mathbf{A}\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0} \iff \mathbf{b} \in \text{cone}(\mathbf{A}).$$

Recap: Comparison of vectors

Componentwise comparisons

For **vectors** $\mathbf{y}, \mathbf{z} \in \mathbb{R}^n$,

$$\mathbf{y} = \mathbf{z} \iff y_i = z_i \text{ for all } i,$$

$$\mathbf{y} \geq \mathbf{z} \iff y_i \geq z_i \text{ for all } i,$$

$$\mathbf{y} > \mathbf{z} \iff y_i \geq z_i \text{ for all } i \text{ and } y_j > z_j \text{ for some } j,$$

$$\mathbf{y} \gg \mathbf{z} \iff y_i > z_i \text{ for all } i.$$

Non-negative orthant

$$\mathbb{R}_+^n \equiv \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} \geq \mathbf{0}\}.$$

Recap: Farkas Lemma

Farkas Lemma

Let $\mathbf{A}$ be an $m \times n$ matrix, $\mathbf{b} \in \mathbb{R}^m$, and

$$F = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{Ax} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}.$$

Exactly one of the following holds:

1. $F \neq \emptyset$;
2. there exists a **vector** $\mathbf{y} \in \mathbb{R}^m$ such that

$$\mathbf{yA} \geq \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} < 0.$$

The second system is called the **Farkas alternative**.

Problem 1: Knowledge used

Knowledge used

- ▶ Cone generated by the columns of a **matrix**:

$$\text{cone}(\mathbf{A}) = \{\mathbf{Ax} : \mathbf{x} \geq \mathbf{0}\}.$$

- ▶ A column that is itself a non-negative linear combination of other columns does not enlarge the cone.
- ▶ Farkas Lemma: exactly one of

$$\mathbf{Ax} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}$$

and

$$\mathbf{yA} \geq \mathbf{0}, \mathbf{y} \cdot \mathbf{b} < 0$$

has a solution.

Problem 1(i): Cone generated by the columns

The columns are

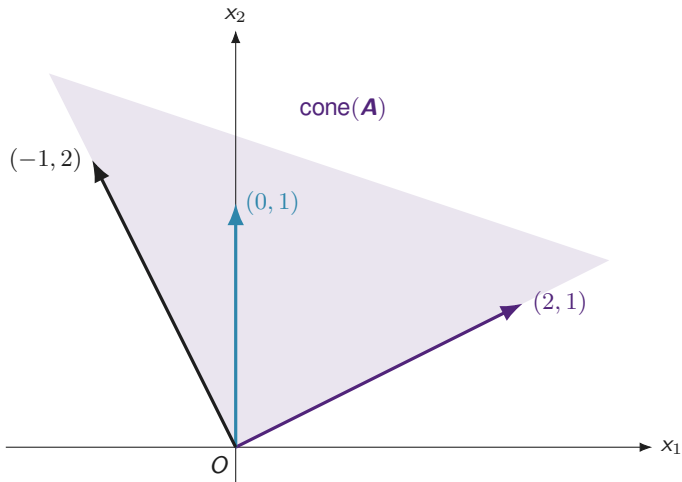
$$\mathbf{a}^1 = (2, 1), \quad \mathbf{a}^2 = (0, 1), \quad \mathbf{a}^3 = (-1, 2).$$

Hence

$$\text{cone}(\mathbf{A}) = \{x_1 \mathbf{a}^1 + x_2 \mathbf{a}^2 + x_3 \mathbf{a}^3 : x_1, x_2, x_3 \geq 0\}.$$

The two extreme rays are generated by $\mathbf{a}^1$ and $\mathbf{a}^3$; the second column lies between them.

Problem 1(i): Sketch



Problem 1(ii): First and third columns

Solution

The cone generated by the first and third columns is the same as the cone generated by all three columns.

Indeed, the second column is a non-negative linear combination of the first and third columns:

$$\frac{1}{5}(2, 1) + \frac{2}{5}(-1, 2) = (0, 1).$$

Therefore,

$$\mathbf{a}^2 \in \text{cone}\{\mathbf{a}^1, \mathbf{a}^3\},$$

and including $\mathbf{a}^2$ does not enlarge the cone.

Problem 1(iii): Farkas alternative

For $\mathbf{b} = (1, 0)$, the Farkas alternative is

$$\mathbf{y}\mathbf{A} \geq \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} < 0.$$

Writing $\mathbf{y} = (y_1, y_2)$,

$$(y_1, y_2) \begin{pmatrix} 2 & 0 & -1 \\ 1 & 1 & 2 \end{pmatrix} \geq (0, 0, 0), \quad y_1 < 0.$$

Thus the alternative is

$$\begin{cases} 2y_1 + y_2 \geq 0, \\ y_2 \geq 0, \\ -y_1 + 2y_2 \geq 0, \\ y_1 < 0. \end{cases}$$

Problem 1(iii): Conclusion

Solution

The feasible region of the Farkas alternative is non-empty. For example,

$$\mathbf{y} = (-1, 2)$$

satisfies

$$2(-1) + 2 = 0, \quad 2 \geq 0, \quad -(-1) + 2(2) = 5 \geq 0,$$

and $y_1 = -1 < 0$.

Hence the Farkas alternative has a solution. By Farkas Lemma, the original system

$$\mathbf{Ax} = \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0}$$

does not have a non-negative solution.

Tutorial Question 2

Problem 2

Use Farkas Lemma to decide whether the system

$$\begin{pmatrix} 4 & 1 & -5 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

has a non-negative solution.

Recap: Farkas Lemma

Farkas Lemma

Let $\mathbf{A}$ be an $m \times n$ matrix, $\mathbf{b} \in \mathbb{R}^m$, and

$$F = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{Ax} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}.$$

Exactly one of the following holds:

1. $F \neq \emptyset$;
2. there exists a **vector** $\mathbf{y} \in \mathbb{R}^m$ such that

$$\mathbf{yA} \geq \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} < 0.$$

The second system is called the **Farkas alternative**.

Problem 2: Knowledge used

Knowledge used

- Farkas Lemma for

$$\mathbf{Ax} = \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0}.$$

- The Farkas alternative is

$$\mathbf{yA} \geq \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} < 0.$$

- To show that the original system has a non-negative solution, show that the Farkas alternative has no solution.

Problem 2: Set up the Farkas alternative

Let

$$\mathbf{A} = \begin{pmatrix} 4 & 1 & -5 \\ 1 & 0 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

For $\mathbf{y} = (y_1, y_2)$, the alternative is

$$(y_1, y_2)\mathbf{A} \geq (0, 0, 0), \quad y_1 + y_2 < 0.$$

Equivalently,

$$\begin{cases} 4y_1 + y_2 \geq 0, \\ y_1 \geq 0, \\ -5y_1 + 2y_2 \geq 0, \\ y_1 + y_2 < 0. \end{cases}$$

Problem 2: Rule out the alternative

Solution

From the second inequality,

$$y_1 \geq 0.$$

Together with $y_1 + y_2 < 0$, this implies

$$y_2 < 0.$$

But then

$$-5y_1 + 2y_2 < 0,$$

which contradicts the third inequality

$$-5y_1 + 2y_2 \geq 0.$$

Therefore, the Farkas alternative has no solution.

By Farkas Lemma, the original system **has a non-negative solution**.

Tutorial Question 3

Problem 3

Let $\mathbf{A}$ be an $m \times n$ **matrix**, and let $\mathbf{b} \in \mathbb{R}^m$. Prove, using Farkas Lemma, that exactly one of the following holds:

1. the system

$$\mathbf{Ax} \geq \mathbf{b}$$

has a non-negative solution $\mathbf{x} \geq \mathbf{0}$;

2. there exists a non-negative **vector** $\mathbf{y} \in \mathbb{R}^m$ such that

$$\mathbf{yA} \leq \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} > 0.$$

Recap: Farkas Lemma

Farkas Lemma

Let $\mathbf{A}$ be an $m \times n$ matrix, $\mathbf{b} \in \mathbb{R}^m$, and

$$F = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{Ax} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}.$$

Exactly one of the following holds:

1. $F \neq \emptyset$;
2. there exists a **vector** $\mathbf{y} \in \mathbb{R}^m$ such that

$$\mathbf{yA} \geq \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} < 0.$$

The second system is called the **Farkas alternative**.

Recap: Comparison of vectors

Componentwise comparisons

For **vectors** $\mathbf{y}, \mathbf{z} \in \mathbb{R}^n$,

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Non-negative orthant

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Recap: Converting linear inequalities

Slack and surplus variables

For a $\leq$ inequality, add a non-negative slack variable:

$$\sum_{j=1}^n a_{ij}x_j \leq b_i \iff \sum_{j=1}^n a_{ij}x_j + s_i = b_i, \quad s_i \geq 0.$$

For a $\geq$ inequality, subtract a non-negative surplus variable:

$$\sum_{j=1}^n a_{ij}x_j \geq b_i \iff \sum_{j=1}^n a_{ij}x_j - s_i = b_i, \quad s_i \geq 0.$$

Problem 3: Knowledge used

Knowledge used

- ▶ A system $\mathbf{Ax} \geq \mathbf{b}$ can be converted to equality form by subtracting a non-negative surplus **vector**.
- ▶ An augmented **matrix** can combine the original variables and auxiliary variables into one equality system.
- ▶ Apply Farkas Lemma to the transformed equality system.
- ▶ A sign change in the Farkas **vector** converts $\mathbf{y}' \leq \mathbf{0}$ into a non-negative **vector**.

Problem 3: Convert the inequalities to equalities

Start from

$$\mathbf{Ax} \geq \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0}.$$

Subtract a non-negative surplus **vector** $\mathbf{s} \geq \mathbf{0}$:

$$\mathbf{Ax} - \mathbf{s} = \mathbf{b}.$$

Using the $m \times m$ identity **matrix** $\mathbf{I}$,

$$\mathbf{Ax} - \mathbf{Is} = \mathbf{b}.$$

Define

$$\mathbf{B} \equiv [\mathbf{A} \mid -\mathbf{I}], \quad \mathbf{z} \equiv (\mathbf{x}, \mathbf{s}).$$

Then the system becomes

$$\mathbf{Bz} = \mathbf{b}, \quad \mathbf{z} \geq \mathbf{0}.$$

Problem 3: Apply Farkas Lemma

Farkas Lemma says that exactly one of

$$Bz = b, \quad z \geq 0$$

and its alternative has a solution.

The alternative is: there exists a **vector** y' such that

$$y'B \geq 0, \quad y' \cdot b < 0.$$

Since $B = [A \mid -I]$,

$$y'B = [y'A \mid -y'] \geq 0.$$

Therefore,

$$y'A \geq 0, \quad y' \leq 0, \quad y' \cdot b < 0.$$

Problem 3: Put the alternative in the required form

Define the non-negative **vector**

$$y \equiv -y' \geq 0.$$

Multiplying the remaining inequalities by -1 gives

$$yA \leq 0$$

and

$$y \cdot b > 0.$$

Hence exactly one of the following holds:

$$\boxed{Ax \geq b, x \geq 0}$$

or

$$\boxed{yA \leq 0, y \geq 0, y \cdot b > 0}.$$

This is the required statement.

Tutorial Question 4

Problem 4

Write down the Farkas alternative for

$$x_1 + 2x_2 + 3x_3 \leq 5,$$

$$x_1 + 3x_2 - 2x_3 \geq 7,$$

$$x_1 + x_2 + x_3 \leq 2,$$

$$x_1 - 2x_2 - 3x_3 \geq 3,$$

with

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0.$$

Recap: Converting linear inequalities

Slack and surplus variables

For a $\leq$ inequality, add a non-negative slack variable:

$$\sum_{j=1}^n a_{ij}x_j \leq b_i \iff \sum_{j=1}^n a_{ij}x_j + s_i = b_i, \quad s_i \geq 0.$$

For a $\geq$ inequality, subtract a non-negative surplus variable:

$$\sum_{j=1}^n a_{ij}x_j \geq b_i \iff \sum_{j=1}^n a_{ij}x_j - s_i = b_i, \quad s_i \geq 0.$$

Recap: Farkas alternative for $\mathbf{Ax} \leq \mathbf{b}$

Primal system

$$\mathbf{Ax} \leq \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0}.$$

Farkas alternative

$$\mathbf{yA} \geq \mathbf{0}, \quad \mathbf{y} \geq \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} < 0$$

Problem 4: Knowledge used

Knowledge used

- ▶ Rewrite all inequalities in a common $\leq$ form.
- ▶ Express the system as

$$\mathbf{Ax} \leq \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0}.$$

- ▶ From Lecture 6, the Farkas alternative for this system is

$$\mathbf{yA} \geq \mathbf{0}, \quad \mathbf{y} \geq \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} < 0.$$

Problem 4: Rewrite the $\geq$ inequalities

The second and fourth inequalities are equivalent to

$$-x_1 - 3x_2 + 2x_3 \leq -7,$$

and

$$-x_1 + 2x_2 + 3x_3 \leq -3.$$

Therefore, with

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix},$$

the system can be written as

$$\mathbf{Ax} \leq \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0}.$$

Problem 4: Matrix form

Here

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -3 & 2 \\ 1 & 1 & 1 \\ -1 & 2 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ -7 \\ 2 \\ -3 \end{pmatrix}.$$

Thus the primal system is

$$\begin{pmatrix} 1 & 2 & 3 \\ -1 & -3 & 2 \\ 1 & 1 & 1 \\ -1 & 2 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \leq \begin{pmatrix} 5 \\ -7 \\ 2 \\ -3 \end{pmatrix}, \quad \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \geq \mathbf{0}.$$

Problem 4: Farkas alternative

Solution

For a **vector** $\mathbf{y} = (y_1, y_2, y_3, y_4)$, the Farkas alternative is

$$\mathbf{yA} \geq \mathbf{0}, \quad \mathbf{y} \geq \mathbf{0}, \quad \mathbf{y} \cdot \mathbf{b} < 0.$$

Therefore,

$$(y_1, y_2, y_3, y_4) \begin{pmatrix} 1 & 2 & 3 \\ -1 & -3 & 2 \\ 1 & 1 & 1 \\ -1 & 2 & 3 \end{pmatrix} \geq (0, 0, 0),$$

$$(y_1, y_2, y_3, y_4) \geq (0, 0, 0, 0),$$

and

$$5y_1 - 7y_2 + 2y_3 - 3y_4 < 0.$$